Complete graph measurements and exact regularity certificates

The selected rational 2-volcano has 46 vertices, 90 directed maps and five levels. All 100 paired problems on every curve have verified rho and BSGS answers, and all 9200 separately counted native replays match answers and all non-clock original statistics. All 9200 independent PARI point equations pass.

For each matched key, BSGS online group-operation counts agree on every curve/generator. Every curve mean is 71837588.1 original group operations. Native non-square Montgomery-multiplication means span 359188416 to 359188419.04, a span of 3.04 out of about 359 million. Palette contrast across that narrow range does not establish a hardness change. Rho curve means span 107895789.39 to 134494184.67 original group operations; its coordinate partition changes under transport. Original and instrumented timings remain separate.

Exact invariant and evidence

For S=Fp[x,y,z], J is generated by separate homogenizations of F,G,H. All 13800 random setup/target systems have reg(J)=26 and reg(S/J)=25; all 920 planted controls have reg(J)=25 and reg(S/J)=24. Each curve has 320 systems: minimum 25, mean 25.9375, maximum 26 for reg(J). These values follow from per-system checked hypotheses, complete-intersection or residual-grid Hilbert-Burch/mapping-cone certificates. PARI independently verifies every actual-field grid and 920 homogeneous polynomial certificates. A separate small-prime genuine-field resolution regression verifies the convention and three structural cases. This is not a claim that Singular computed a resolution at the large characteristic, or that the affine unit ideal has regularity 26.

The original top-forms Hilbert cutoff 13, temporary accumulator degree 23 and pair sugar 26 retain their separate definitions. The prior unsupported-resolution pilot and all failed builds are preserved. Certificate verification uses a new frozen protocol and timing boundary; it does not reduce the recorded original GB cost.

Unresolved requirement

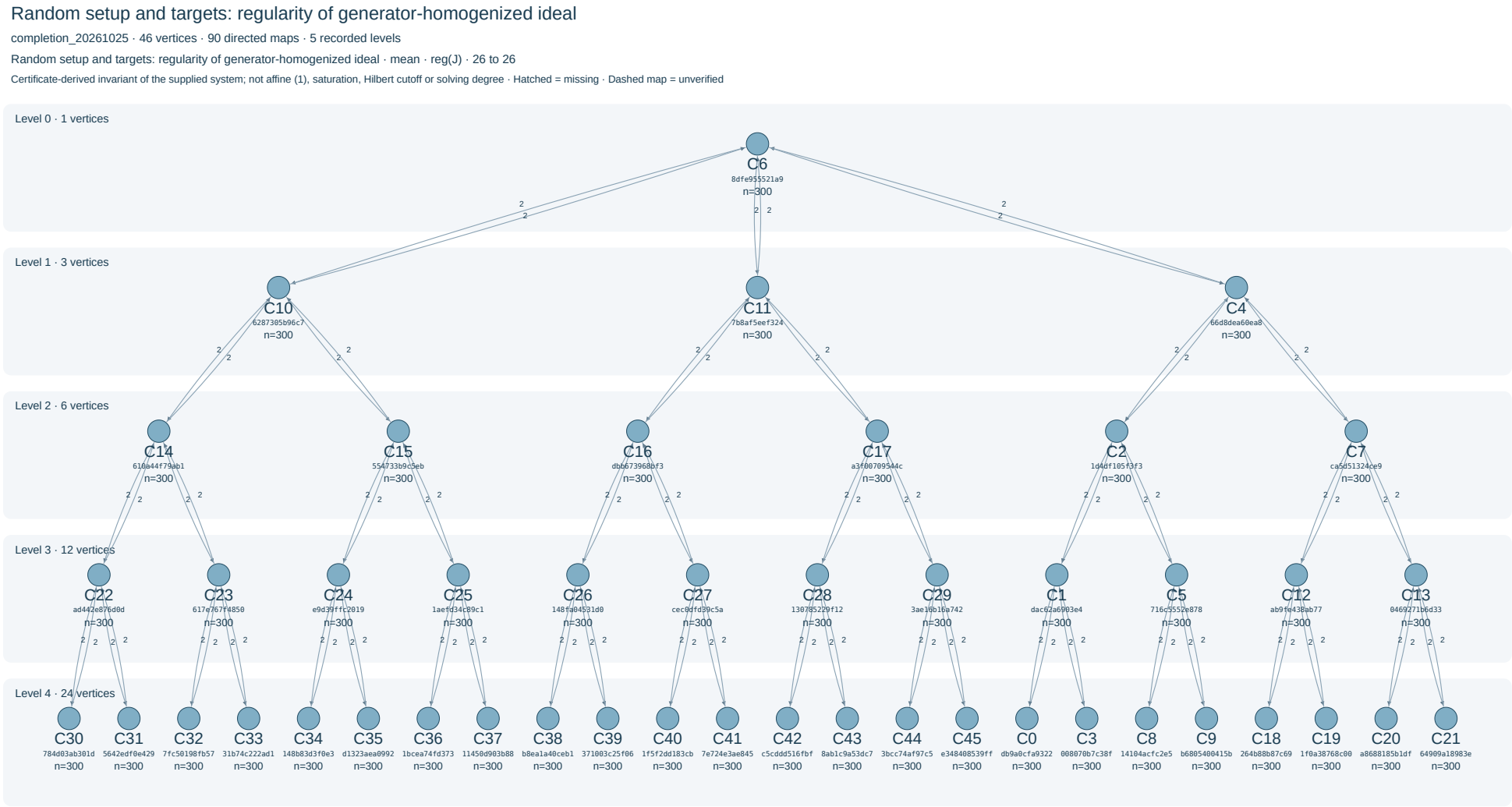
The frozen strict two-sum IC candidate has zero random relations, setup rank zero and zero random-corpus recoveries. Successful full IC recovery remains blocked by missing useful relation collection and descent. Positive controls are not recoveries, and bounded failure is not an impossibility theorem. The entire requested experiment is not declared complete.

Master table: per_curve.csv. Complete per-generator statistics, per-point cross-curve summaries, numeric query projections, setup/destruction and all retained native attempts are in tables/. Native multiplication and squaring calls are disjoint; per-case M+S is a derived total. Reductions overlap conversions/products. These are source-level primitive calls, not CPU instructions. Timings are descriptive phase data, without an admitted common-cost/fastest comparison. Public projections omit every scalar witness and raw RNG word; original records remain local.

Sources: FIELD_COUNTER_SEMANTICS_CORRECTION.md, source/regularity_cert.cpp, REGULARITY_CERT_PROTOCOL.md, REGULARITY_CERT_FROZEN.json, REGULARITY_CERTIFICATE_PLAN.md, source/completion_report.mjs and independent_regularity_audit.json. Original source/binary/input identities remain unchanged.

Random setup and targets: regularity of generator-homogenized ideal

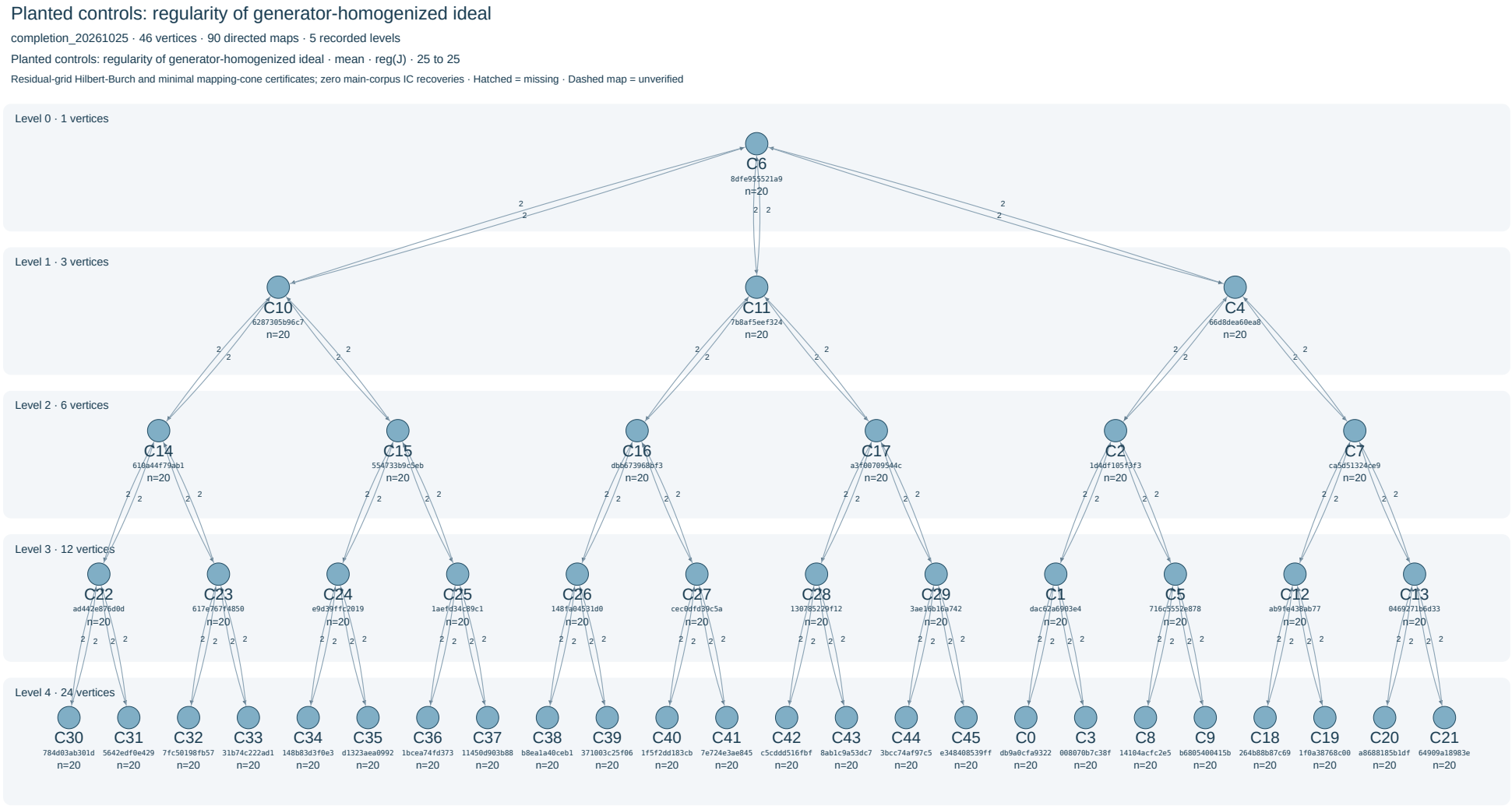
Certificate-derived invariant of the supplied system; not affine (1), saturation, Hilbert cutoff or solving degree. Unit reg(J).



Full exact identities, map evidence and measurements: graph_data.json / interactive inspector.

Planted controls: regularity of generator-homogenized ideal

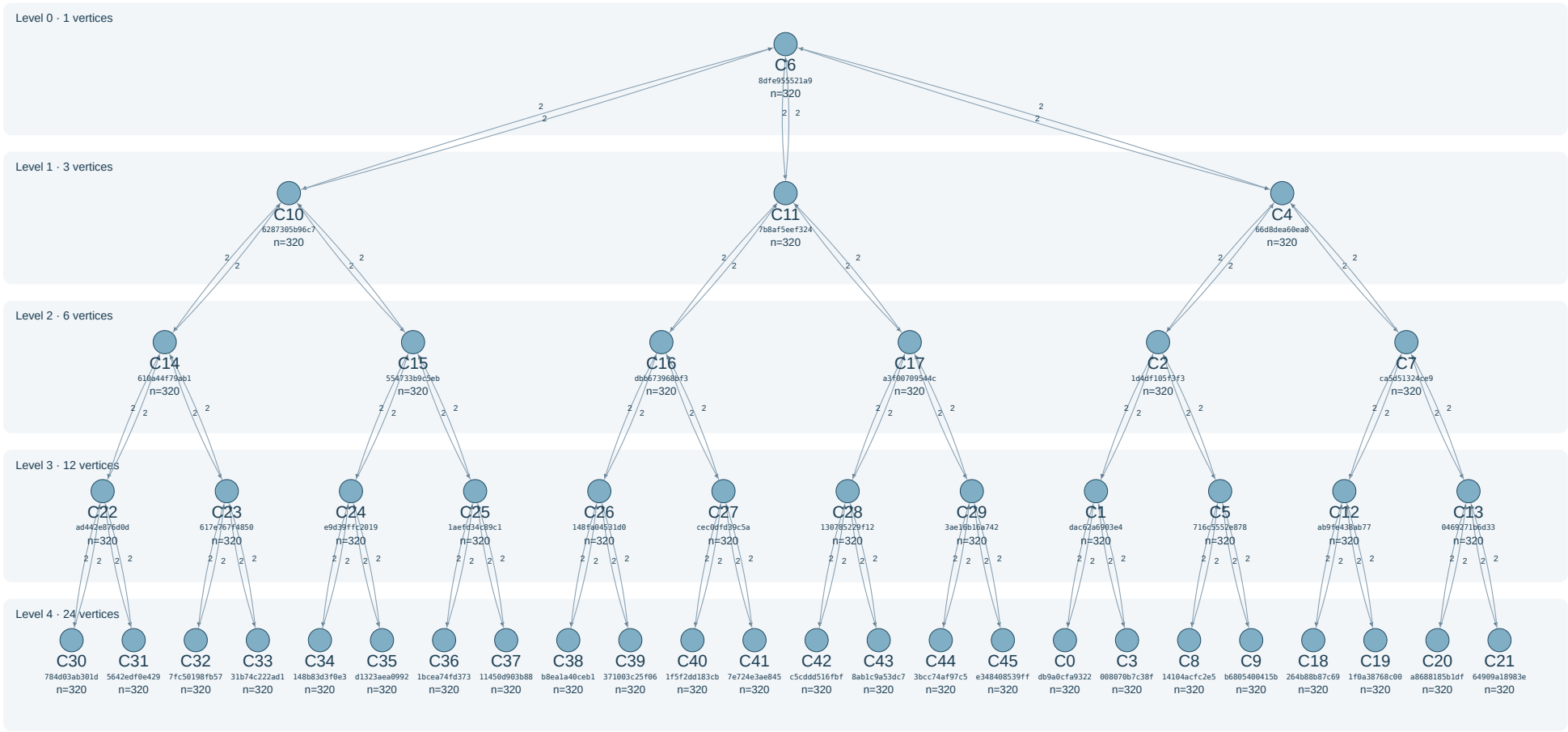
Residual-grid Hilbert-Burch and minimal mapping-cone certificates; zero main-corpus IC recoveries. Unit reg(J).



Full exact identities, map evidence and measurements: graph_data.json / interactive inspector.

Regularity certificate verification: counted field products

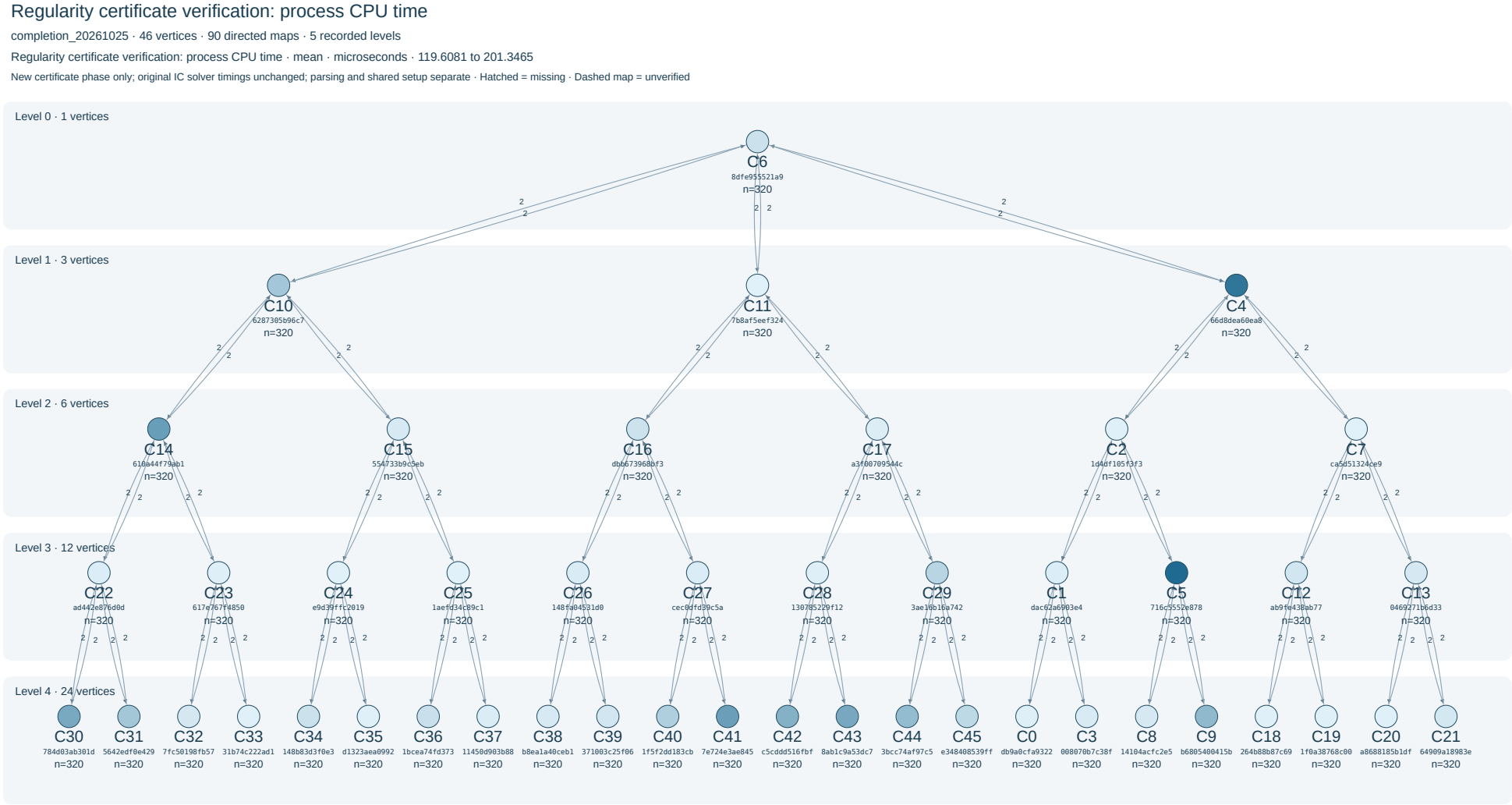
New C++ proof-verification phase; includes full grid and actual polynomial identity checks; shared setup separate. Unit Fp multiplications including inverse exponentiation.



Full exact identities, map evidence and measurements: graph_data.json / interactive inspector.

Regularity certificate verification: process CPU time

New certificate phase only; original IC solver timings unchanged; parsing and shared setup separate. Unit microseconds.



Full exact identities, map evidence and measurements: graph_data.json / interactive inspector.